Microsoft 365

ICE-3207 Software and Requirement Engineering

Nazneen Akhter

Assistant Professor

Department of ICT, FST

Software Development Life Cycle

Planning
↓
Requirement Analysis
↓
Design
↓
Implementation
↓
Testing
↓
Deployment
↓
Maintenance
↺



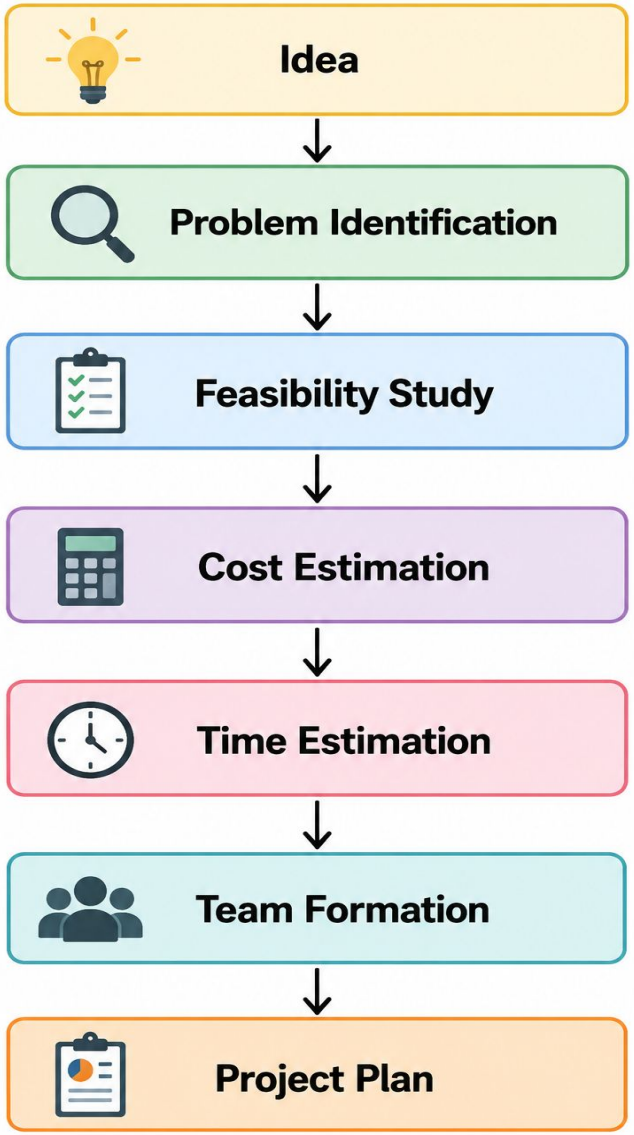
Step1–Planning

Library Management System (LMS)

A university library currently records book borrowing manually in notebooks. Students often wait in long lines, books are misplaced, and librarians cannot quickly determine whether a book is available.

What problem will be solved?
Is the project worth doing?
How much money is required?
How long will it take?
Who will develop it?

Steps of Planning



Planning Step	Question Answered	Library Management System Example
Idea	What is our idea?	We need software to automate the university library and replace the manual notebook-based system.
Problem Identification	What problem are we trying to solve?	Students wait in long lines, books are misplaced, and librarians cannot quickly check book availability.
Feasibility Study	Is the project practical?	Can we build it with available technology? Can the university afford it? Will librarians and students use it? Can it be completed on time? Does it comply with university policies?
Cost Estimation	How much money is needed?	Estimate the budget for software development, hardware, database, training, testing, and deployment.
Time Estimation	How long will the project take?	Estimate the duration for planning, requirements, design, development, testing, and deployment (e.g., 14 weeks).
Team Formation	Who will develop the project?	Form a team consisting of a Project Manager, Business Analyst, UI/UX Designer, Software Developers, Database Administrator, QA Engineer, and System Administrator.
Project Plan	What is the final roadmap?	Prepare a project plan describing the project objectives, scope, budget, timeline, team members, risks, and expected deliverables.